SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

E – Commerce Platform for Personalized Retail Experiences

Group No: 25-26J-526

Checklist 01

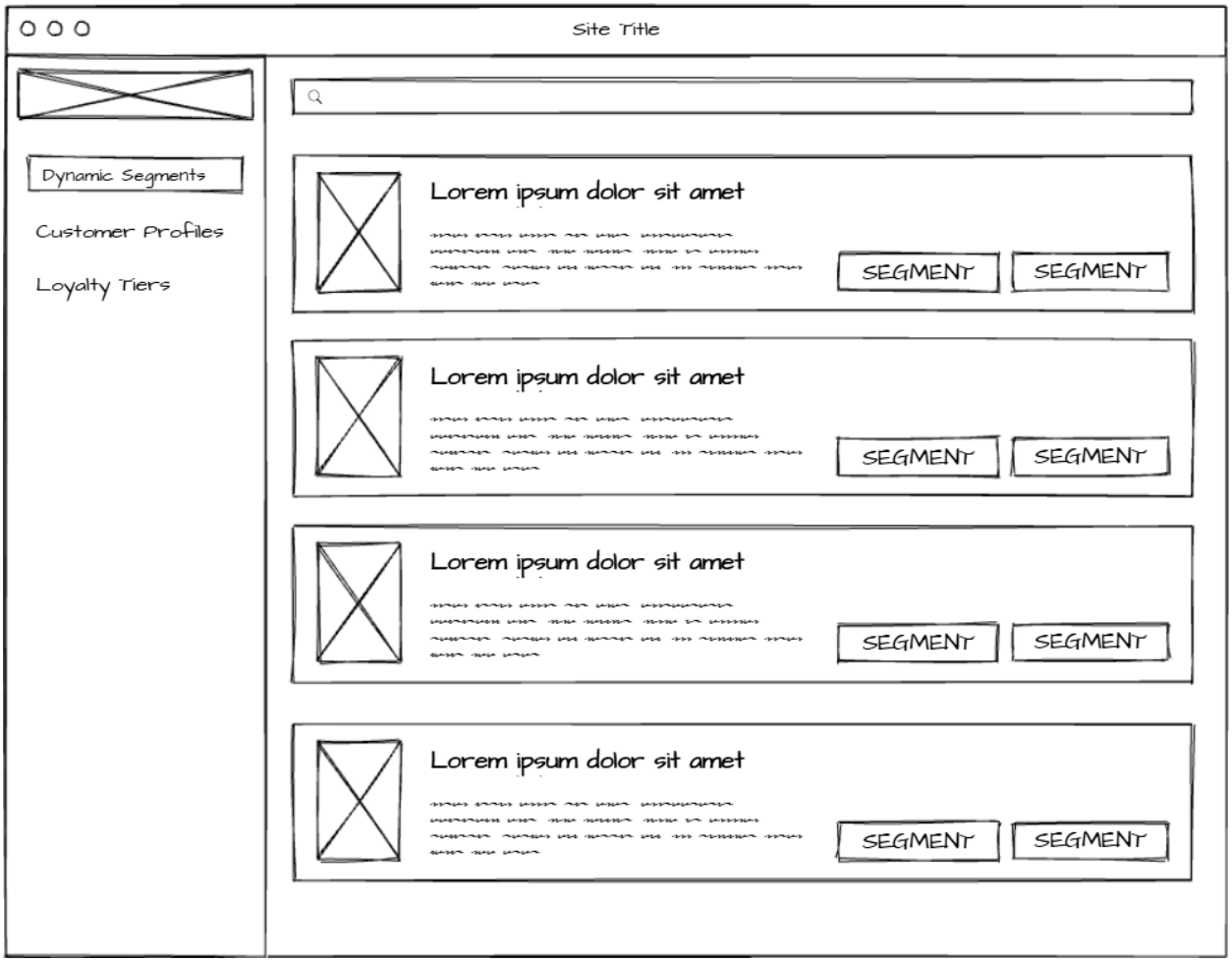
11/01/2026

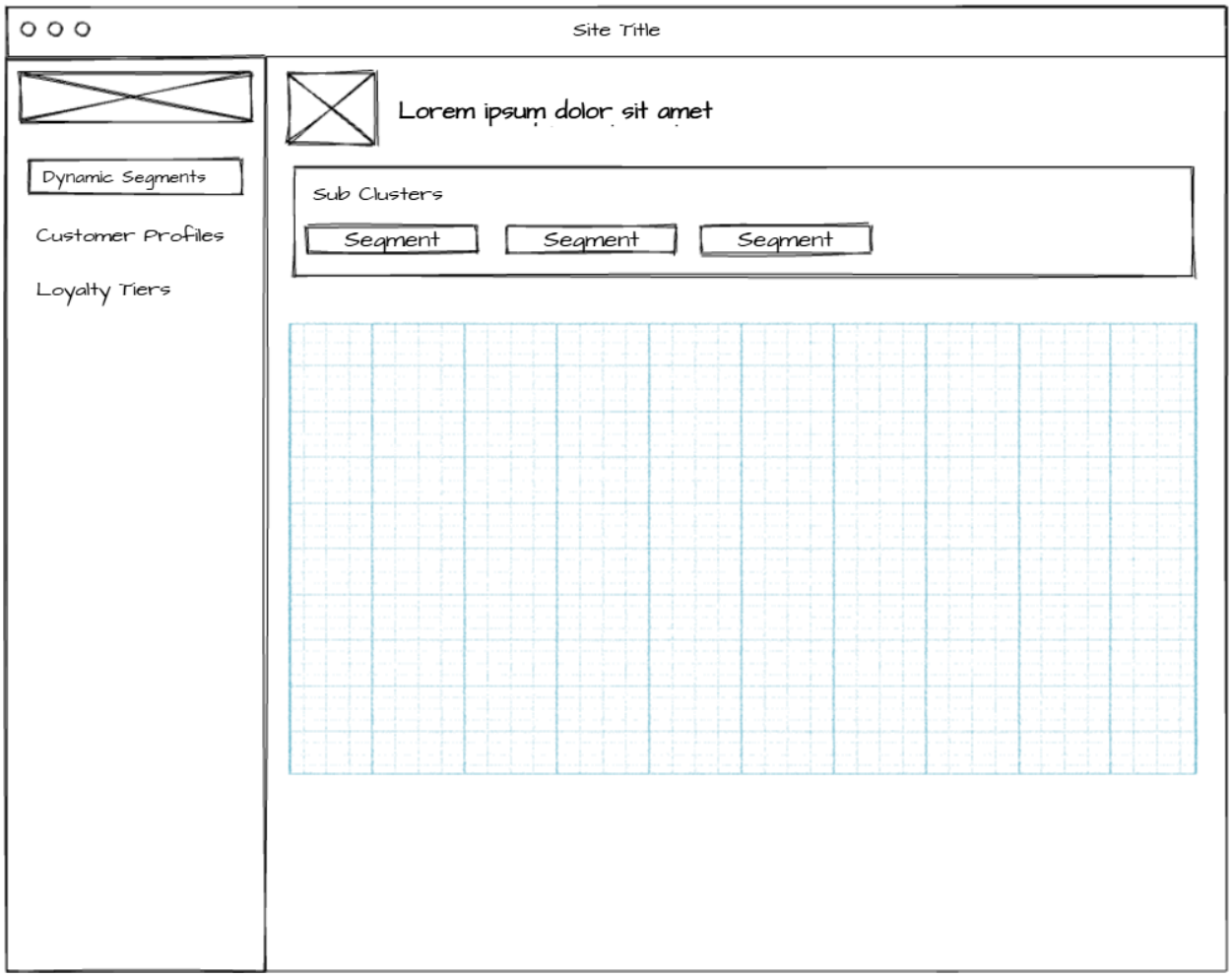
Part A – GitHub Repository

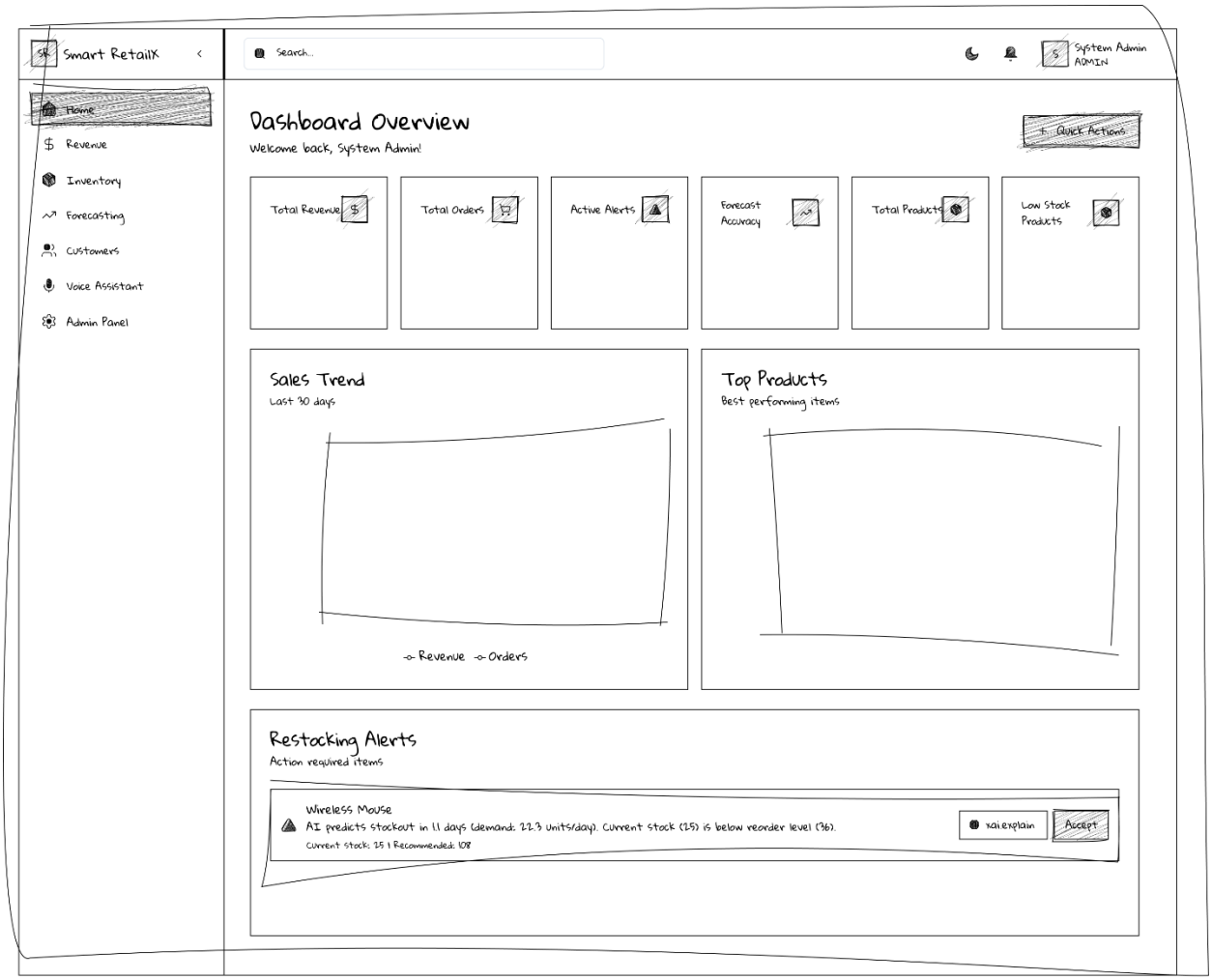
GitHub Repository: SmartRetailX/SmartRetailX

Part B – Design & Mockups

Sketches




Site Title
Dynamic Segments
Customer Profiles
Loyalty Tiers
Lorem ipsum dolor sit amet
Sub Clusters
Segment
Segment
Segment


Smart RetailX
Search...
System Admin
ADMIN
Home
Revenue
Inventory
Forecasting
Customers
Voice Assistant
Admin Panel
Dashboard Overview
Welcome back, System Admin!
+ Quick Actions
Total Revenue
Total Orders
Active Alerts
Forecast Accuracy
Total Products
Low Stock Products
Sales Trend
Last 30 days
Revenue
Orders
Top Products
Best performing items
Restocking Alerts
Action required items
Wireless Mouse
AI predicts stockout in 11 days (demand: 22.3 units/day). Current stock (25) is below reorder level (36).
Current stock: 25 | Recommended: 108
xai explain
Accept

AI Restock Explanation

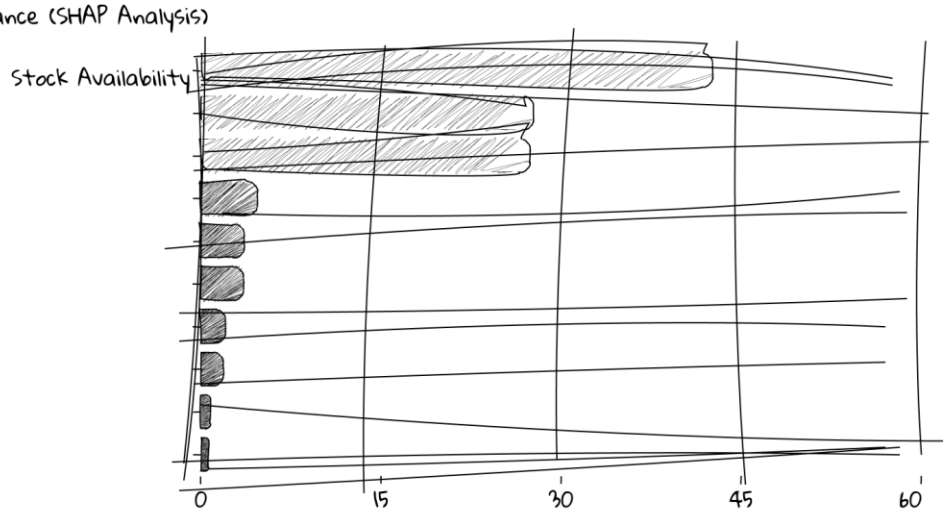
Understanding why AI recommends this restock action

Product

Key Metrics

Predicted Demand	Current Stock	Baseline Demand	Days Until Stockout	Recommended	Reorder Level

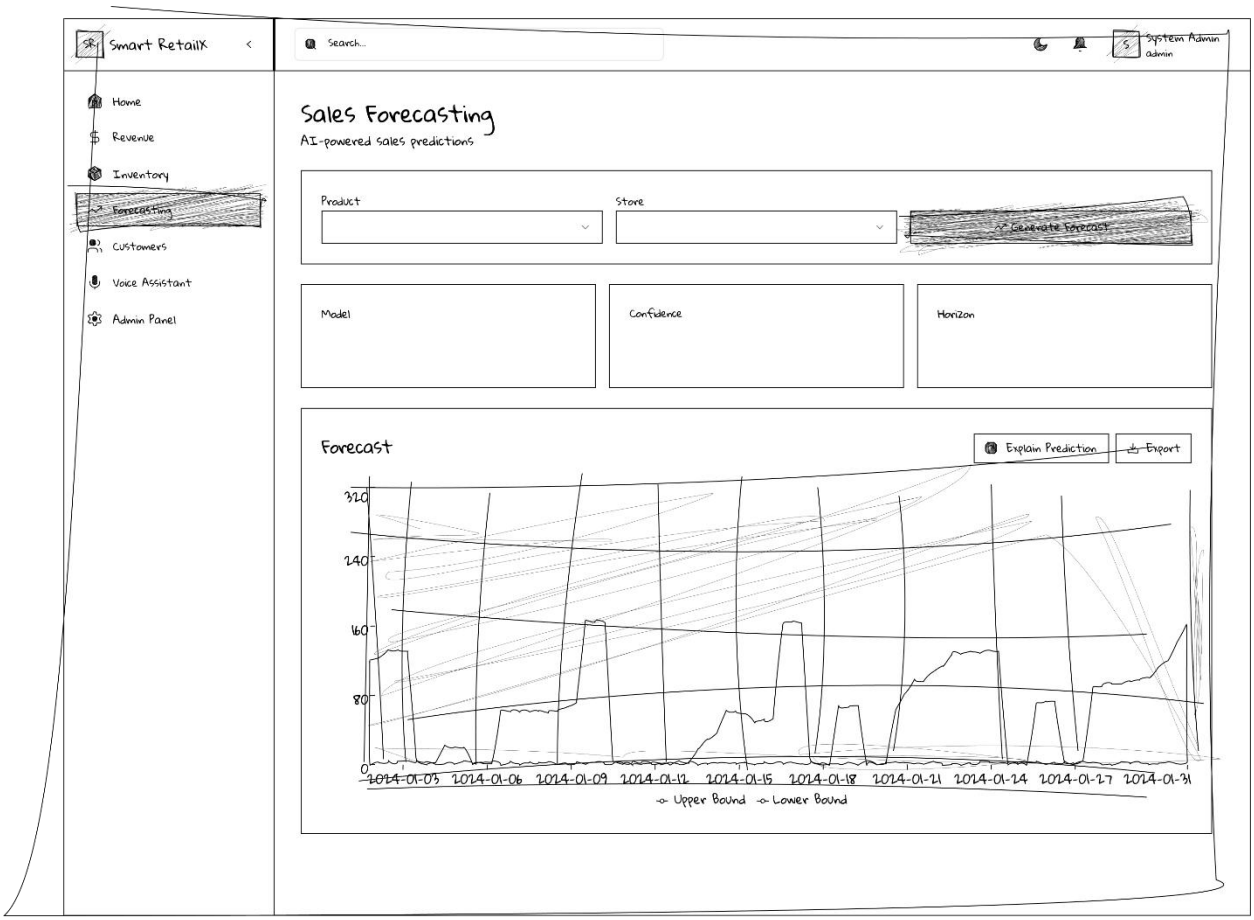
Feature Importance (SHAP Analysis)

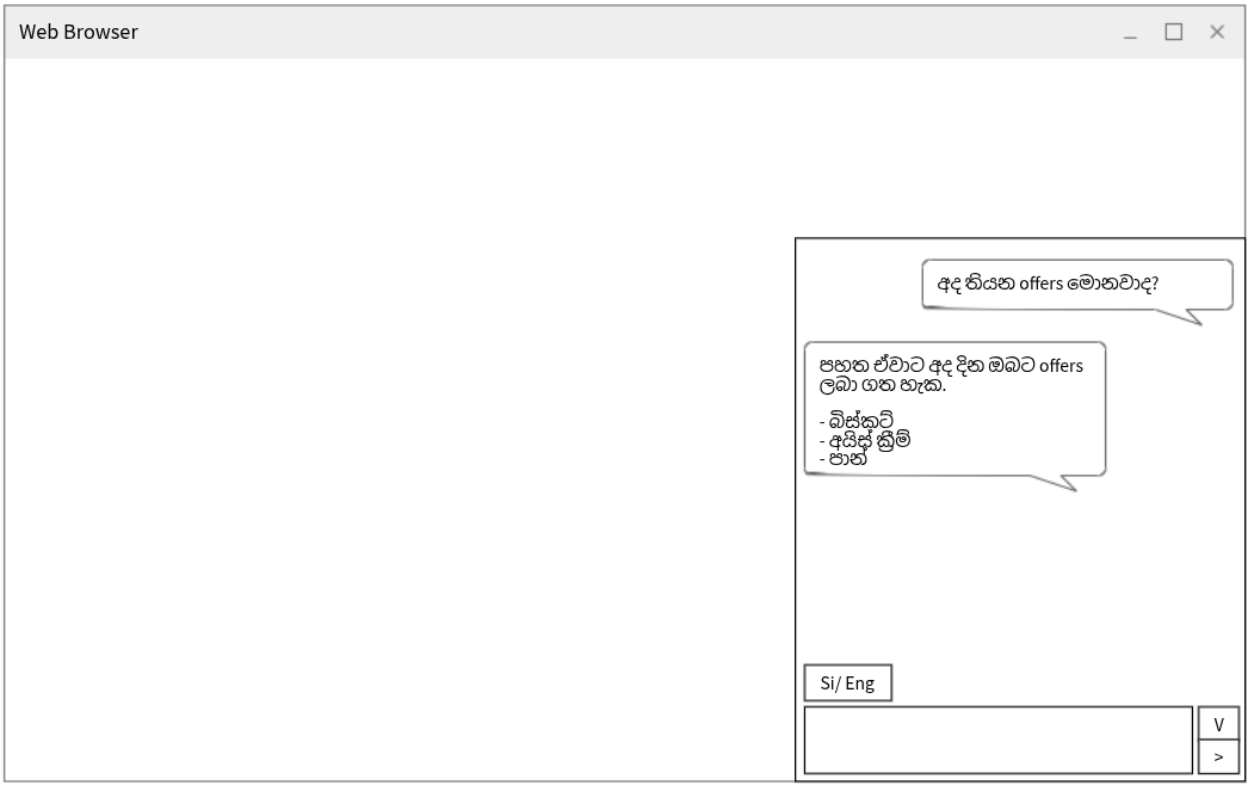


Detailed Feature Analysis

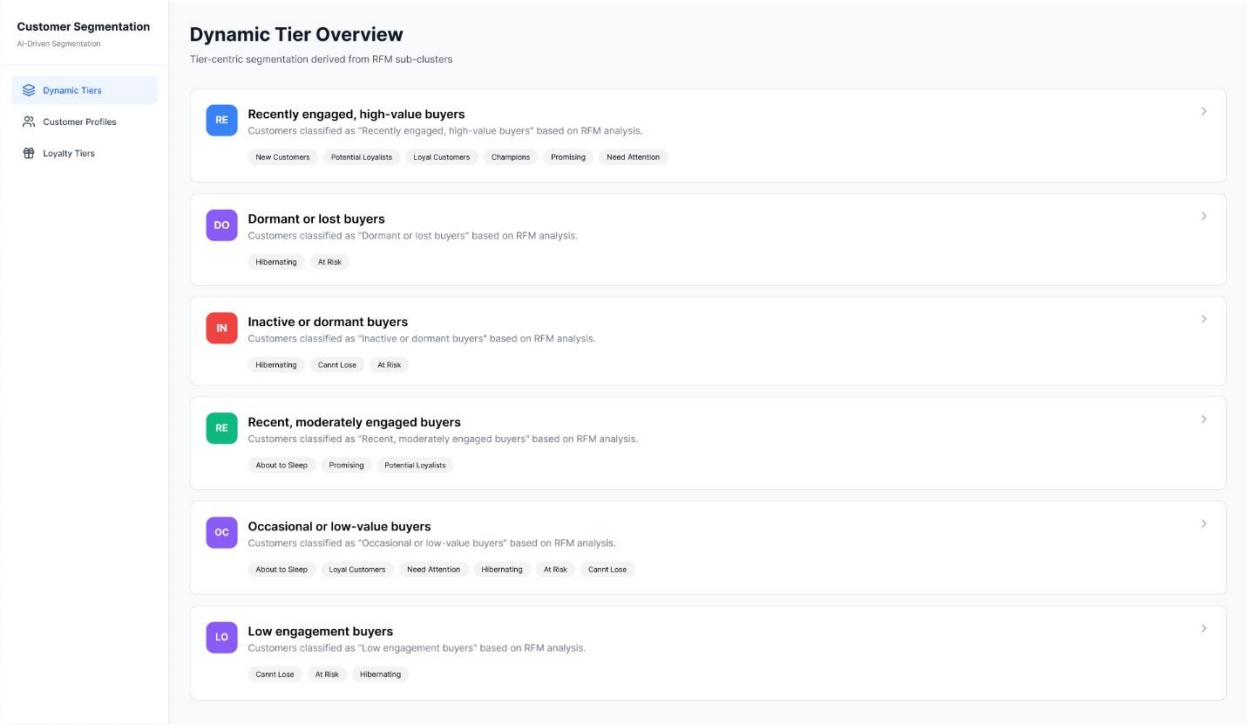
#1 Stock Availability 42.62 Units

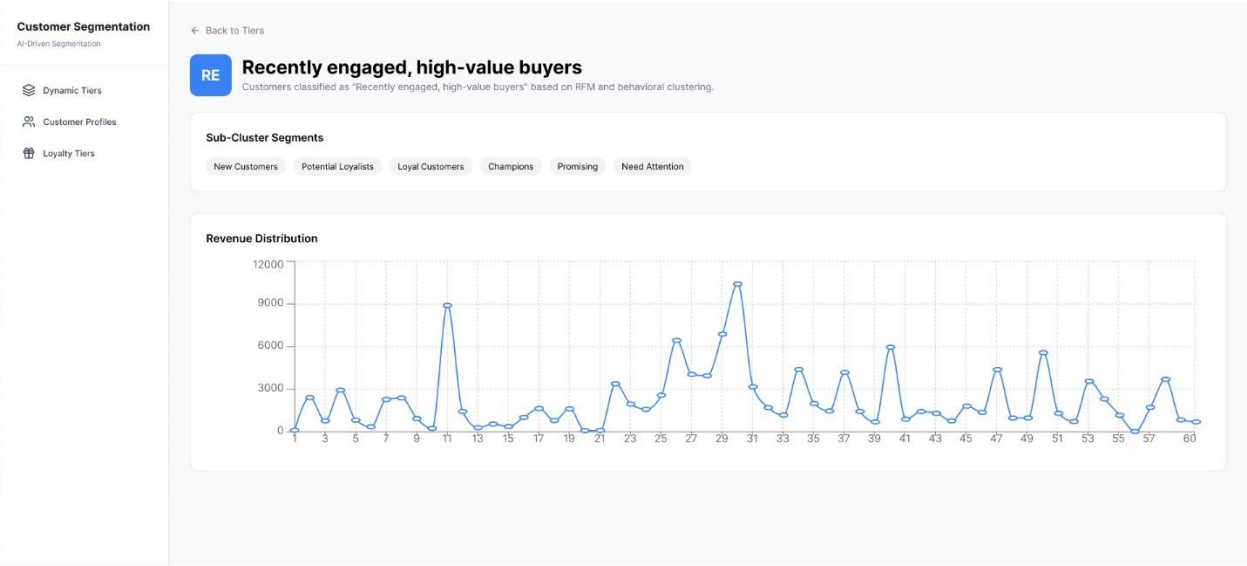
Value: 167 Impact: 42.62 Importance40.3%


Smart RetailX
Search...
System Admin
admin
Home
Revenue
Inventory
Forecasting
Customers
Voice Assistant
Admin Panel
Sales Forecasting
AI-powered sales predictions
Product
Store
Generate Forecast
Model
Confidence
Horizon
Forecast
Explain Prediction
Export
320
240
160
80
0
2024-01-03
2024-01-06
2024-01-09
2024-01-12
2024-01-15
2024-01-18
2024-01-21
2024-01-24
2024-01-27
2024-01-31
Upper Bound
Lower Bound


Web Browser
අද තියන offers මොනවාද?
පහත ඒවාට අද දින ඔබට offers ලබා ගත හැක.
- බිස්කට්
- අයිස් ක්‍රීම්
- පාන්
Si/ Eng
V
>

UIs Before Feedback





SR Smart RetailX

Admin | Customer

SA System Admin
ADMIN

Dashboard Overview

Welcome back, System Admin!

Total Revenue	Total Orders	Active Alerts	Forecast Accuracy	Total Products
LKR 1,250,000.00	4,567	3	94.2%	20
↑ 12.5% vs last period	↑ 8.3% vs last period	↓ 15.2% vs last period	↑ 2.1% vs last period	0.0% vs last period

Create Promotion

Select Product

Choose a product...

Discount Percentage: 10%

5% 25% 50%

Promotion Type

Personalized

Number of Available Discounts

50

Send Promotion

Target Customers (AI Predicted)

Select a product to see target customers

AI will predict customers with highest buying probability

SR Smart RetailX

Admin | Customer

DS Dinesh Silva
CUSTOMER

My Promotions

Hello Dinesh Silva! Check out your exclusive offers

Available Promotions

2 offers available

Chips Premium

Lays

LKR 722.93

~~LKR 850.59~~

Redeem

Valid 7 days

Coffee Instant

Nescafe

LKR 1000.00

~~LKR 1250.00~~

-45 left!

✓ Redeemed

Valid 7 days

Smart RetailX

Search...

System Admin
ADMIN

- Home
- Revenue
- Inventory
- Forecasting
- Customers
- Voice Assistant
- Admin Panel

Dashboard Overview

Welcome back, System Admin!

Quick Actions

Total Revenue	Total Orders	Active Alerts	Forecast Accuracy	Total Products	Low Stock Products
LKR 1,250,000.00	4,567	3	94.2%	20	3
↑ 12.5% vs last period	↑ 8.3% vs last period	↓ 15.2% vs last period	↑ 2.1% vs last period	0.0% vs last period	0.0% vs last period

Sales Trend

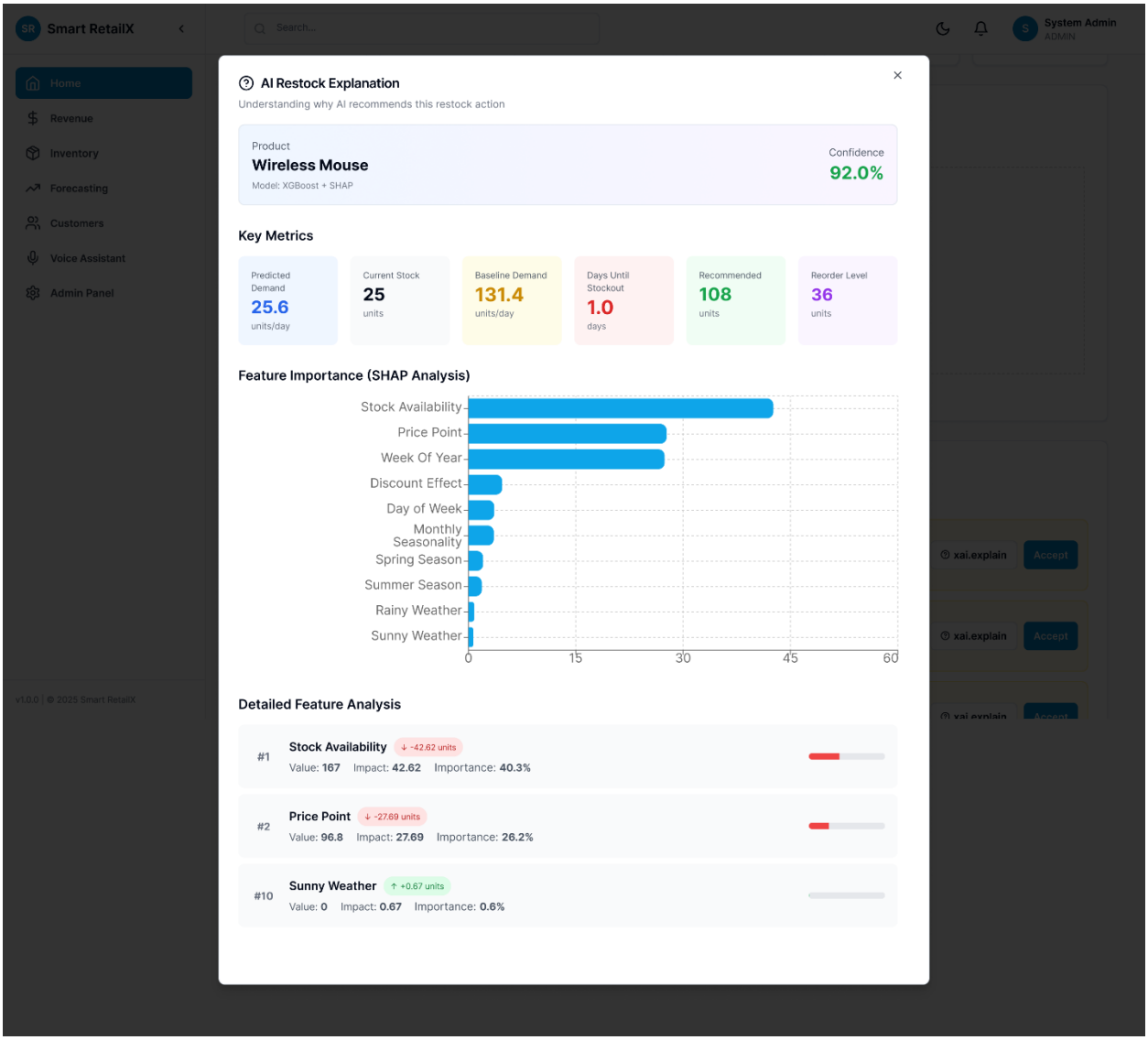
Last 30 days

Revenue Orders

Top Products

Best performing items

v1.0.0 | © 2025 Smart RetailX


Smart RetailX
Search...
System Admin
ADMIN
Home
Revenue
Inventory
Forecasting
Customers
Voice Assistant
Admin Panel
v1.0.0 | © 2025 Smart RetailX
AI Restock Explanation
Understanding why AI recommends this restock action
Product
Wireless Mouse
Model: XGBoost + SHAP
Confidence
92.0%
Key Metrics
Predicted Demand
25.6
units/day
Current Stock
25
units
Baseline Demand
131.4
units/day
Days Until Stockout
1.0
days
Recommended
108
units
Reorder Level
36
units
Feature Importance (SHAP Analysis)
Stock Availability
Price Point
Week Of Year
Discount Effect
Day of Week
Monthly Seasonality
Spring Season
Summer Season
Rainy Weather
Sunny Weather
0
15
30
45
60
Detailed Feature Analysis
#1
Stock Availability
-42.62 units
Value: 167 Impact: 42.62 Importance: 40.3%
#2
Price Point
-27.69 units
Value: 96.8 Impact: 27.69 Importance: 26.2%
#10
Sunny Weather
+0.67 units
Value: 0 Impact: 0.67 Importance: 0.6%
xai.explain
Accept

Smart RetailX

Search...

System Admin
admin

- Home
- Revenue
- Inventory
- Forecasting
- Customers
- Voice Assistant
- Admin Panel

Sales Forecasting

AI-powered sales predictions

Product: Sports Jacket (P0014)

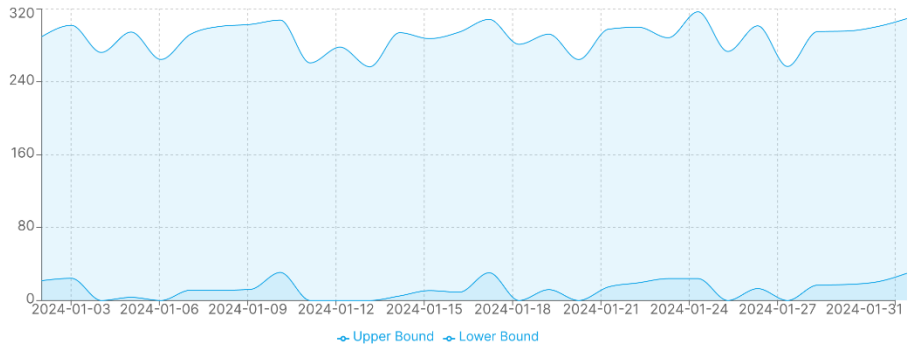
Store: Colombo Central (S001)

Generate Forecast

Model	Confidence	Horizon
Prophet	87.0%	30 days

Forecast

Explain Prediction | Export



Key Drivers

Day of Week Effect Weekend sales are 12% lower	12% Impact
Promotions & Holidays Sales decrease by 5% during promotions	6% Impact
Discount Effect Higher discounts boost sales by 1%	1% Impact

v1.0.0 | © 2025 Smart RetailX

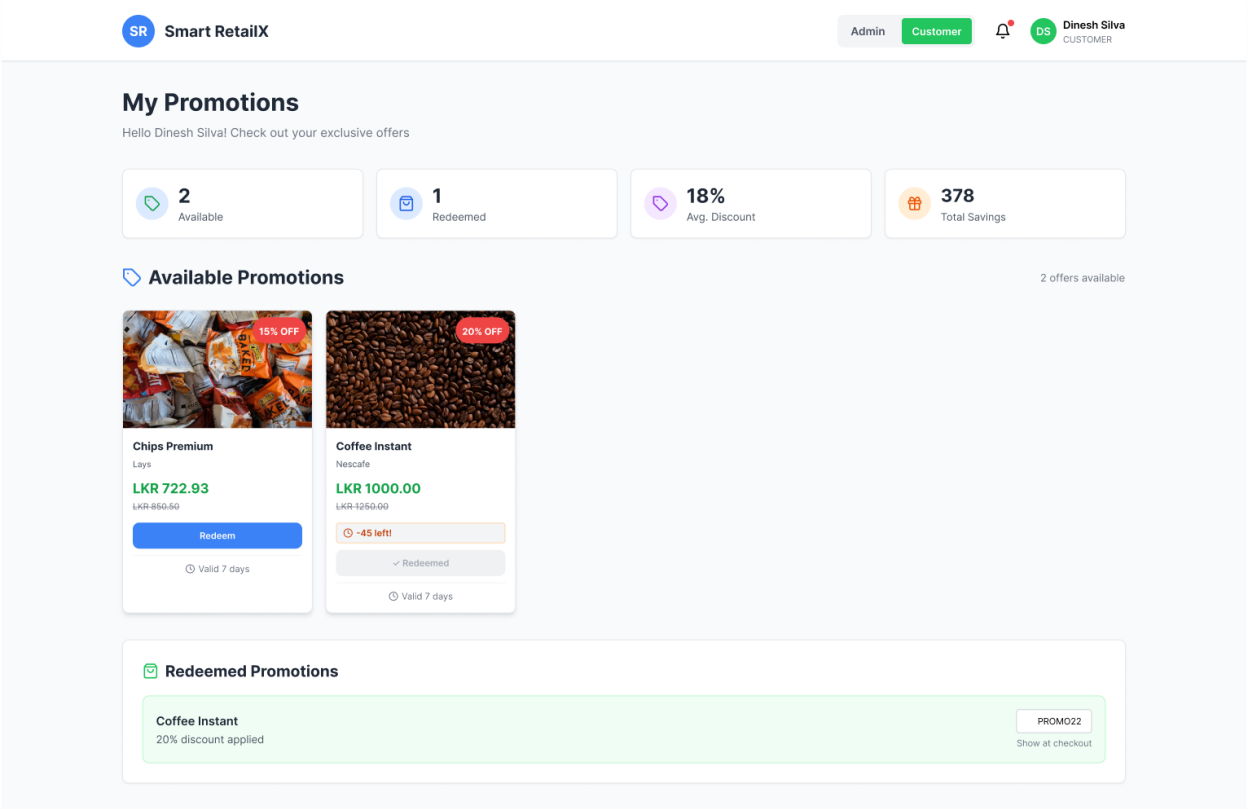
User Feedback:

1. The earlier UI was visually clean but lacked immediate insight. While the tier names and labels were clear, it did not show how many customers belonged to each tier, their revenue contribution, or their associated risk, making it difficult to understand the relative importance of each segment at a glance. The updated UI addresses this by presenting key metrics such as customer count, revenue, behavior score, and churn risk directly within each tier, along with an overall summary at the top, enabling faster understanding, clearer comparison, and more informed decision-making.
2. Need an option to send bulk promotions also with personalized promotions. In current promotion system, only personalized promotions are allowed
3. There should be an option to see the current active promotions including product, type, discount amount, price, promotion usage and promotion status
4. In client-side promotion view, client should be able to see redeemed promotions with product names and discount amounts
5. The prediction numbers are helpful, but I have to figure out how they connect. For example, when I see '25.6 units/day' and 'stockout in 1.0 days,' I must do the math myself that the stock is too low. It would be clearer if there was a short description explaining this in plain language right next to the numbers
6. Switching between English and Sinhala is important for our team, but having to go into the settings menu to change language is inconvenient. It would be much better if there was a language switch button at the top of the screen where it's easy to find and use quickly.

Improvements of UI:

1. Added option to send bulk promotions with UI enhancement for the bulk promotion option
2. Added current active promotion section in the admin view with requested data
3. Enhanced client promotion view with redeemed promotion view section
4. Added descriptive explanatory text that connects key metrics in plain language.
5. Moved the language toggle from the settings menu to the main header.

UI s After Feedback



SR **Smart RetailX**

Admin | Customer

SA **System Admin**
ADMIN

Dashboard Overview

Welcome back, System Admin!

Total Revenue	Total Orders	Active Alerts	Forecast Accuracy	Total Products
LKR 1,250,000.00	**4,567**	**3**	**94.2%**	**20**
↑ 12.5% vs last period	↑ 8.3% vs last period	↓ 15.2% vs last period	↑ 2.1% vs last period	0.0% vs last period

Create Promotion

Select Product

Tea Bags - Dilmah (LKR 890.00)

Tea Bags
Category: Beverages
Brand: Dilmah
LKR 890.00

Discount Percentage: 10%

5% 25% 50%

Final Price:
LKR 801.00
Save LKR 89.00

Promotion Type

Personalized | Bulk

Number of Available Discounts

50

Send Promotion

Target Customers (AI Predicted)

Top 5 Customers Selected
Based on purchase history and ML predictions

1. **Lakshmi Rathnayake**
lakshmi.r@email.com
Gampaha | Age: 41
Buy Probability **84.1%**

2. **Sandya Mendis**
sandya.m@email.com
Kandy | Age: 31
Buy Probability **53.5%**

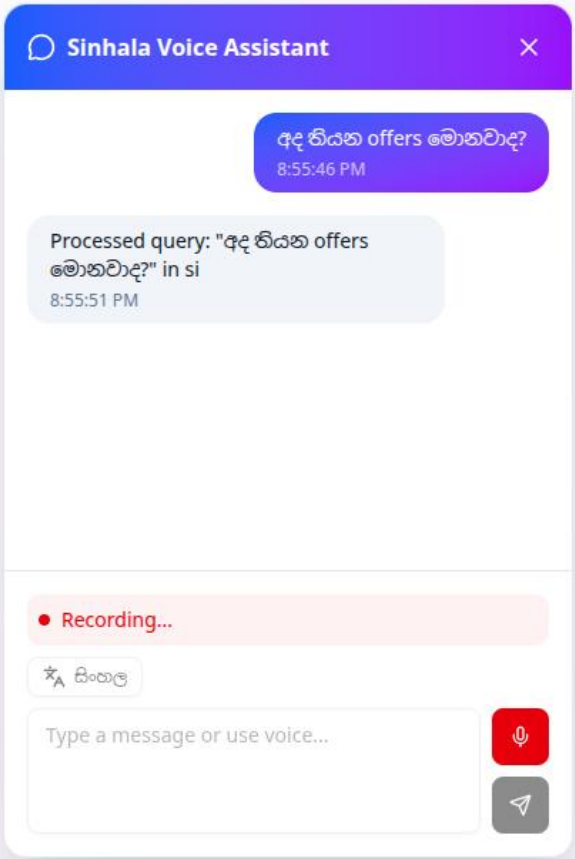
3. **Priya Jayawardena**
priya.j@email.com
Colombo | Age: 44
Buy Probability **51.8%**

4. **Sunil Fernando**
sunil.f@email.com
Kurunegala | Age: 50
Buy Probability **51.5%**

5. **Chandrika Dias**
chandrika.d@email.com
Kurunegala | Age: 40
Buy Probability **49.0%**

Active Promotions

PRODUCT	TYPE	DISCOUNT	PRICE	USAGE	STATUS
Chips Premium Lays • Snacks	personalized	15% OFF	~~LKR 850.50~~ **LKR 722.93**	12 / 50	active
Coffee Instant Nescafe • Beverages	bulk	20% OFF	~~LKR 1250.00~~ **LKR 1000.00**	45 redeemed	active
Rice 5kg Araliya • Grains	personalized	10% OFF	~~LKR 1450.00~~ **LKR 1305.00**	8 / 30	active
Buns 2 Maliban • Bakery	personalized	10% OFF	~~LKR 713.62~~ **LKR 642.26**	0 / 50	active
Bread 3 Anchor • Bakery	personalized	10% OFF	~~LKR 1154.43~~ **LKR 1038.99**	0 / 50	active


Sinhala Voice Assistant
අද තියන offers මොනවාද?
8:55:46 PM
Processed query: "අද තියන offers මොනවාද?" in si
8:55:51 PM
Recording...
සිංහල
Type a message or use voice...

Customer Segmentation
AI-Driven Segmentation

- Dynamic Tiers
- Customer Profiles
- Loyalty Tiers

Dynamic Tier Overview

Tier-centric segmentation derived from RFM sub-clusters

Active Tiers	Total Customers	Total Revenue	Avg Score
6	141	$236,312.21	82.8

RE **Recently engaged, high-value buyers**
Customers classified as "Recently engaged, high-value buyers" based on RFM analysis.

Customers	Revenue	Behavior Score	Churn Risk
60	$133,144.21	185.0	6.6%

New Customers · Potential Loyalists · Loyal Customers · Champions · Promising · Need Attention

DO **Dormant or lost buyers**
Customers classified as "Dormant or lost buyers" based on RFM analysis.

Customers	Revenue	Behavior Score	Churn Risk
15	$12,744.41	33.3	81.9%

Hibernating · At Risk

IN **Inactive or dormant buyers**
Customers classified as "Inactive or dormant buyers" based on RFM analysis.

Customers	Revenue	Behavior Score	Churn Risk
13	$17,049.38	56.9	62.2%

Hibernating · Cannt Lose · At Risk

RE **Recent, moderately engaged buyers**
Customers classified as "Recent, moderately engaged buyers" based on RFM analysis.

Customers	Revenue	Behavior Score	Churn Risk
5	$1,858.36	68.0	14.2%

About to Sleep · Promising · Potential Loyalists

OC **Occasional or low-value buyers**
Customers classified as "Occasional or low-value buyers" based on RFM analysis.

Customers	Revenue	Behavior Score	Churn Risk
24	$39,826.96	89.3	26.5%

About to Sleep · Loyal Customers · Need Attention · Hibernating · At Risk · Cannt Lose

LO **Low engagement buyers**
Customers classified as "Low engagement buyers" based on RFM analysis.

Customers	Revenue	Behavior Score	Churn Risk
24	$31,688.89	64.2	43.8%

Cannt Lose · At Risk · Hibernating

Customer Segmentation
AI-Driven Segmentation

- Dynamic Tiers
- Customer Profiles
- Loyalty Tiers

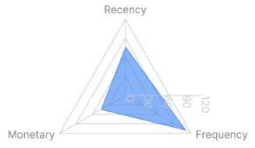
← Back to Tiers

RE Recently engaged, high-value buyers

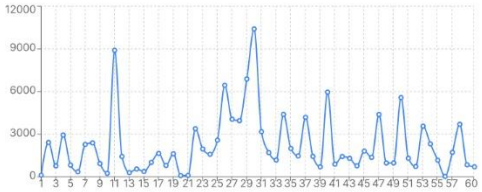
Customers classified as "Recently engaged, high-value buyers" based on RFM and behavioral clustering.

Customers	Total Revenue	Behavior Score	Churn Risk
60	$133,144.21	185.0	6.6%

Behavioral Profile



Revenue Distribution



Sub-Cluster Segments

New Customers · Potential Loyalists · Loyal Customers · Champions · Promising · Need Attention

Smart RetailX

Search...

System Admin
ADMIN

- Home
- Revenue
- Inventory
- Forecasting
- Customers
- Voice Assistant
- Admin Panel

Dashboard Overview

Welcome back, System Admin!

+ Quick Actions

Total Revenue	Total Orders	Active Alerts	Forecast Accuracy	Total Products	Low Stock Products
LKR 1,250,000.00	4,567	3	94.2%	20	3
↑ 12.5% vs last period	↑ 8.3% vs last period	↓ 15.2% vs last period	↑ 2.1% vs last period	0.0% vs last period	0.0% vs last period

Sales Trend

Last 30 days

Revenue Orders

Top Products

Best performing items

Restocking Alerts

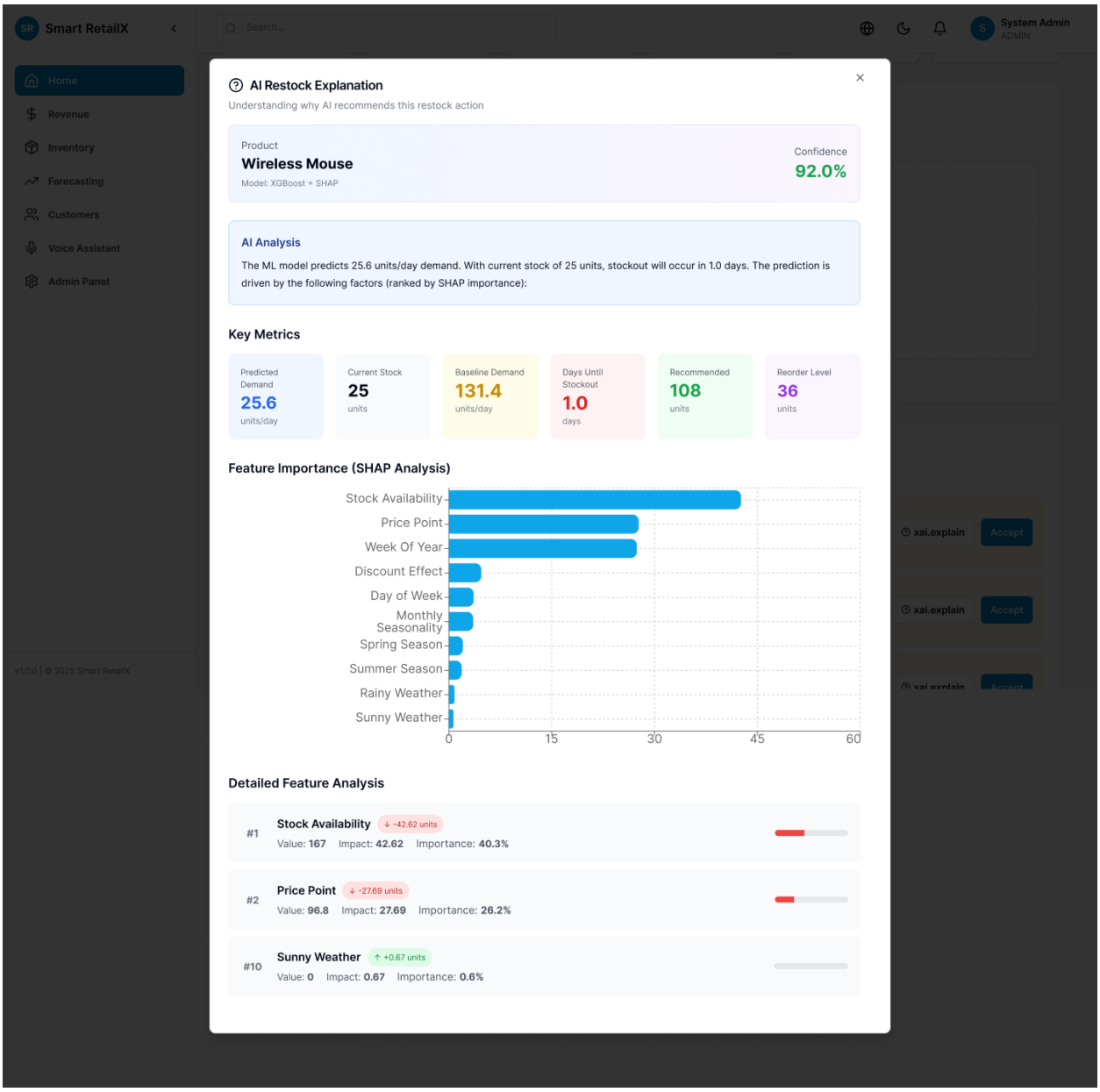
Action required items

Wireless Mouse
AI predicts stockout in 1.1 days (demand: 22.3 units/day). Current stock (25) is below reorder level (36).
Current stock: 25 | Recommended: 108
xai.explain Accept

Bluetooth Speaker
AI predicts stockout in 0.9 days (demand: 16.4 units/day). Current stock (15) is below reorder level (28).
Current stock: 15 | Recommended: 84
xai.explain Accept

Building Blocks Set
AI predicts stockout in 0.5 days (demand: 16.0 units/day). Current stock (8) is below reorder level (22).
Current stock: 8 | Recommended: 66
xai.explain Accept

v1.0.0 | © 2025 Smart RetailX


Smart RetailX
Search...
System Admin
ADMIN
Home
Revenue
Inventory
Forecasting
Customers
Voice Assistant
Admin Panel
v1.0.0 | © 2025 Smart RetailX
AI Restock Explanation
Understanding why AI recommends this restock action
Product
Wireless Mouse
Model: XGBoost + SHAP
Confidence
92.0%
AI Analysis
The ML model predicts 25.6 units/day demand. With current stock of 25 units, stockout will occur in 1.0 days. The prediction is driven by the following factors (ranked by SHAP importance):
Key Metrics
Predicted Demand
25.6
units/day
Current Stock
25
units
Baseline Demand
131.4
units/day
Days Until Stockout
1.0
days
Recommended
108
units
Reorder Level
36
units
Feature Importance (SHAP Analysis)
Stock Availability
Price Point
Week Of Year
Discount Effect
Day of Week
Monthly Seasonality
Spring Season
Summer Season
Rainy Weather
Sunny Weather
0
15
30
45
60
Detailed Feature Analysis
#1
Stock Availability
-42.62 units
Value: 167 Impact: 42.62 Importance: 40.3%
#2
Price Point
-27.69 units
Value: 96.8 Impact: 27.69 Importance: 26.2%
#10
Sunny Weather
+0.67 units
Value: 0 Impact: 0.67 Importance: 0.6%
xai.explain
Accept